```
TC-06 [Condition Coverage]
  Description → Condition fill<=0 → TRUE; fill>100 → FALSE; OR → TRUE
  Input       → bin='BIN001', fill=-10.0, zone='ZONE_A'
  Expected    → ERROR: Invalid fill level
  Actual      → ERROR: Invalid fill level
  Status      → ✅ PASS

TC-07 [Condition Coverage]
  Description → Condition fill<=0 → FALSE; fill>100 → TRUE; OR → TRUE
  Input       → bin='BIN001', fill=105.0, zone='ZONE_A'
  Expected    → ERROR: Invalid fill level
  Actual      → ERROR: Invalid fill level
  Status      → ✅ PASS

TC-08 [Condition Coverage]
  Description → fill>=90 → FALSE; fill>=70 → TRUE; STANDARD assigned
  Input       → bin='BIN002', fill=75.0, zone='ZONE_A'
  Expected    → ROUTE_STANDARD
  Actual      → ROUTE_STANDARD
  Status      → ✅ PASS

TC-09 [Condition Coverage]
  Description → fill>=90 → FALSE; fill>=70 → FALSE; SCHEDULED assigned
  Input       → bin='BIN004', fill=50.0, zone='ZONE_A'
  Expected    → ROUTE_SCHEDULED
  Actual      → ROUTE_SCHEDULED
  Status      → ✅ PASS

TC-10 [Path Coverage]
  Description → Full valid path at minimum boundary: 1%
  Input       → bin='BIN001', fill=1.0, zone='ZONE_A'
  Expected    → ROUTE_SCHEDULED
  Actual      → ROUTE_SCHEDULED
  Status      → ✅ PASS

TC-11 [Path Coverage]
  Description → Full valid path at maximum boundary: 100%, ZONE_B
  Input       → bin='BIN001', fill=100.0, zone='ZONE_B'
  Expected    → ROUTE_PRIORITY_2
  Actual      → ROUTE_PRIORITY_2
  Status      → ✅ PASS

TC-12 [Loop Coverage — 0 iterations]
  Description → Loop executes 0 iterations (immediate exit choice)
  Input       → bin=None, fill=None, zone=None
  Expected    → Loop 0 times
  Actual      → Loop 0 times
  Status      → ✅ PASS

TC-13 [Loop Coverage — 1 iteration]
  Description → Loop executes exactly 1 iteration then exits
  Input       → bin='BIN001', fill=85.0, zone='ZONE_A'
  Expected    → ROUTE_STANDARD
  Actual      → ROUTE_STANDARD
  Status      → ✅ PASS

TC-14 [Loop Coverage — 3 iterations]
  Description → Loop executes 3 iterations; last result PRIORITY_1
  Input       → bin='BIN002', fill=95.0, zone='ZONE_A'
  Expected    → ROUTE_PRIORITY_1
  Actual      → ROUTE_PRIORITY_1
  Status      → ✅ PASS

TC-15 [Branch + Condition Coverage]
  Description → fill=1 (boundary): fill<=0→F, fill>100→F, fill>=90→F, fill>=70→F → SCHEDULED
  Input       → bin='BIN001', fill=1.0, zone='ZONE_A'
  Expected    → ROUTE_SCHEDULED
  Actual      → ROUTE_SCHEDULED
  Status      → ✅ PASS

──────────────────────────────────────────
Results: 15 PASSED | 0 FAILED | 15 TOTAL
Overall Coverage : 100.0%

Coverage by Type:
  Statement Coverage                  2/2  (100%)
  Branch Coverage                     3/3  (100%)
  Condition Coverage                  4/4  (100%)
  Path Coverage                       2/2  (100%)
  Loop Coverage                       3/3  (100%)
  Branch + Condition Coverage         1/1  (100%)
──────────────────────────────────────────
```

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TC-11: BVA: Fill Level = 0 (just below min)
  Input  → citizen='CIT001', bin='BIN001', fill=0.0
  Expected → INVALID – at least 1
  Actual   → Fill level must be at least 1%.
  Status   → ✅ PASS

TC-12: BVA: Fill Level = 1 (minimum valid boundary)
  Input  → citizen='CIT001', bin='BIN001', fill=1.0
  Expected → VALID – Report submitted
  Actual   → Report submitted successfully. Report ID: RPT_TEST_TC-12
  Status   → ✅ PASS

TC-13: BVA: Fill Level = 100 (maximum valid boundary)
  Input  → citizen='CIT001', bin='BIN001', fill=100.0
  Expected → VALID – Report submitted
  Actual   → Report submitted successfully. Report ID: RPT_TEST_TC-13
  Status   → ✅ PASS

TC-14: BVA: Fill Level = 101 (just above max)
  Input  → citizen='CIT001', bin='BIN001', fill=101.0
  Expected → INVALID – exceed
  Actual   → Fill level cannot exceed 100%.
  Status   → ✅ PASS

TC-15: BVA: Fill Level = 99 (just below max, all fields)
  Input  → citizen='CIT002', bin='BIN002', fill=99.0
  Expected → VALID – Report submitted
  Actual   → Report submitted successfully. Report ID: RPT_TEST_TC-15
  Status   → ✅ PASS

────────────────────────────────────────
Results: 15 PASSED | 0 FAILED | 15 TOTAL
────────────────────────────────────────

1. Raise Report (Manual Input)
2. Run Automated Black Box Test Cases (15)
3. Run Automated White Box Test Cases (15)
4. Exit

Enter choice: 3

====================================================================
WHITE BOX TEST CASES – Statement / Branch / Condition / Path / Loop
Method Under Test: assign_route() & handle_report_overflow()
====================================================================

TC-01 [Statement Coverage]
  Description → All statements in assign_route() executed on valid mid-range fill
  Input       → bin='BIN001', fill=85.0, zone='ZONE_A'
  Expected    → ROUTE_STANDARD
  Actual      → ROUTE_STANDARD
  Status      → ✅ PASS

TC-02 [Statement Coverage]
  Description → Error-return statement executed for negative fill
  Input       → bin='BIN001', fill=-5.0, zone='ZONE_A'
  Expected    → ERROR: Invalid fill level
  Actual      → ERROR: Invalid fill level
  Status      → ✅ PASS

TC-03 [Branch Coverage]
  Description → fill>=90 TRUE branch AND zone==ZONE_A TRUE branch
  Input       → bin='BIN003', fill=95.0, zone='ZONE_A'
  Expected    → ROUTE_PRIORITY_1
  Actual      → ROUTE_PRIORITY_1
  Status      → ✅ PASS

TC-04 [Branch Coverage]
  Description → fill>=90 TRUE branch AND zone!=ZONE_A FALSE branch
  Input       → bin='BIN003', fill=95.0, zone='ZONE_B'
  Expected    → ROUTE_PRIORITY_2
  Actual      → ROUTE_PRIORITY_2
  Status      → ✅ PASS

TC-05 [Branch Coverage]
  Description → fill<=0 condition TRUE – error branch
  Input       → bin='BIN001', fill=0.0, zone='ZONE_A'
  Expected    → ERROR: Invalid fill level
  Actual      → ERROR: Invalid fill level
  Status      → ✅ PASS
```

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================================================================
  SMART WASTE MANAGEMENT SYSTEM - Report Overflowing Bin
================================================================

  1. Raise Report (Manual Input)
  2. Run Automated Black Box Test Cases (15)
  3. Run Automated White Box Test Cases (15)
  4. Exit

  Enter choice: 2

================================================================
  BLACK BOX TEST CASES - ECP & BVA
  Method: handle_report_overflow() / ValidationManager
================================================================

  TC-01: ECP Valid: All fields correct, fill=85%
    Input  → citizen='CIT001', bin='BIN001', fill=85.0
    Expected → VALID - Report submitted
    Actual   → Report submitted successfully. Report ID: RPT_TEST_TC-01
    Status   → ✅ PASS

  TC-02: ECP Valid: Minimum valid fill (1%)
    Input  → citizen='CIT002', bin='BIN002', fill=1.0
    Expected → VALID - Report submitted
    Actual   → Report submitted successfully. Report ID: RPT_TEST_TC-02
    Status   → ✅ PASS

  TC-03: ECP Valid: Maximum valid fill (100%)
    Input  → citizen='CIT001', bin='BIN003', fill=100.0
    Expected → VALID - Report submitted
    Actual   → Report submitted successfully. Report ID: RPT_TEST_TC-03
    Status   → ✅ PASS

  TC-04: ECP Valid: Different bin and citizen
    Input  → citizen='CIT003', bin='BIN004', fill=60.0
    Expected → VALID - Report submitted
    Actual   → Report submitted successfully. Report ID: RPT_TEST_TC-04
    Status   → ✅ PASS

  TC-05: ECP Valid: Photo optional (no photo)
    Input  → citizen='CIT001', bin='BIN005', fill=72.0
    Expected → VALID - Report submitted
    Actual   → Report submitted successfully. Report ID: RPT_TEST_TC-05
    Status   → ✅ PASS

  TC-06: ECP Invalid: Bin ID empty
    Input  → citizen='CIT001', bin='', fill=80.0
    Expected → INVALID - mandatory
    Actual   → Bin ID is mandatory.
    Status   → ✅ PASS

  TC-07: ECP Invalid: Bin ID not in system
    Input  → citizen='CIT001', bin='BIN999', fill=90.0
    Expected → INVALID - does not exist
    Actual   → Bin ID 'BIN999' does not exist in the system.
    Status   → ✅ PASS

  TC-08: ECP Invalid: Citizen ID empty
    Input  → citizen='', bin='BIN001', fill=70.0
    Expected → INVALID - mandatory
    Actual   → Citizen ID is mandatory.
    Status   → ✅ PASS

  TC-09: ECP Invalid: Fill level empty/None
    Input  → citizen='CIT001', bin='BIN001', fill=None
    Expected → INVALID - numeric
    Actual   → Fill level must be a numeric value.
    Status   → ✅ PASS

  TC-10: ECP Invalid: Fill level negative (-10%)
    Input  → citizen='CIT001', bin='BIN001', fill=-10.0
    Expected → INVALID - negative
    Actual   → Fill level cannot be negative.
    Status   → ✅ PASS

  TC-11: BVA: Fill Level = 0 (just below min)
    Input  → citizen='CIT001', bin='BIN001', fill=0.0
    Expected → INVALID - at least 1
```

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